Two blocks are incompatible with the critical head

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses: $n = 2r \geq 18$, $G \leq \mathrm{Alt}(\Omega)$ is transitive, its string C-marking has connected noncommutation diagram, and every full maximal omission is intransitive. The actual first three marks form the acquired six-point head on D. The acquired early-cut calculation gives

$$\tau = \rho_3 = (6\ b)\delta, \quad \sigma = \rho_4, \quad x = b^\sigma \neq b, \quad \delta b = b, \quad \delta x = x,$$

where δ is odd and supported on $E = \Omega \setminus D$. The group $K = G_{\geq 5}$ fixes $D \cup \{b\}$ and centralizes δ; $B = G_{\geq 4}$ fixes D pointwise.

Lemma 1 (A saturated forest)**.** *Let $T = \langle t_1, \ldots, t_{N-2} \rangle$ act faithfully on N points. Assume the t_i are nonidentity involutions, the noncommutation diagram on the distinguished marks is connected, T is intransitive, and deleting each t_i changes its orbit partition. Then T fixes one point and is the full natural S_{N-1} on the others. Every distinguished mark is one transposition there.*

Proof. For each i select an edge $a_i -_i a_i^{t_i}$ whose endpoints lie in different orbits of the omission of t_i. These $N-2$ edges form a forest: a cycle would give an avoiding path for one of its selected edges. Thus the selected forest has exactly two components. Both lie in T-orbits. Since T is intransitive, these components are precisely its two orbits.

Deleting a selected edge j splits its tree into two pieces. The actual omission of t_j contains all the other selected edges, and cannot join these two pieces by eligibility. Hence it preserves each piece, as well as the other tree component. Every t_i with $i \neq j$ preserves this cut. In a tree the only two distinct vertices not separated by any cut other than cut i are the endpoints of edge i. It follows that t_i is exactly its selected transposition. The transpositions of each tree generate the full symmetric group on that tree. If both trees had edges, their two nonempty collections of marks would commute with one another, disconnecting the diagram. Thus one tree is a singleton, as claimed. □

Theorem 2. *Under the standing hypotheses, B cannot be transitive on E and preserve a partition of E into two blocks of equal size.*

Proof. Suppose that the blocks are U, V, each of size $m = (n-6)/2 = r - 3 \geq 6$, with $b \in U$. Every mark of K fixes b, so it preserves both blocks. Transitivity forces σ to interchange them. Being an involution, σ is a matching of all m points of U with all m points of V. It is even and fixes D, so m is even. In particular $x \in V$.

Put $e = \rho_5$ and $H = G_{\geq 6}$. Every element of H preserves U, V, centralizes σ, and fixes b, x. Identifying matched pairs by $u \leftrightarrow u^\sigma$, H acts diagonally on

$$U_0 = U \setminus \{b\}, \qquad V_0 = V \setminus \{x\}.$$

Its action T on U_0 is faithful: a kernel element fixes U, then fixes V by commutation with σ, and fixes D already. Consequently the inherited string C-marking has rank $r - 6 = m - 3 = N - 2$ on $N = m - 1$ points. Every mark is nonidentity, and its noncommutation diagram is connected, being a final interval of the original connected string.

Each row omission changes the row orbit partition. Otherwise its identical diagonal action on V_0, together with the fixed points b, x, D, would give the same point-orbit partition as H. Adding

the other distinguished generators would then give the same orbit partition with or without the omitted generator: orbits of a generated group are the equivalence relation generated by the individual orbit relations. The full ambient omission would be transitive, contrary to hypothesis.

The mark e moves x. Indeed, if it fixed x, then $\langle e, H\rangle$ would fix both b, x, while σ interchanges them, leaving $\{b, x\}$ invariant under B. This contradicts transitivity since $|E| > 2$. Write $y = x^e \in V_0$. Since δ centralizes e and fixes x, it fixes y. It also centralizes H and fixes b, x.

If T is transitive, the accepted complete transitive-rank theorem (L047, Theorem 1.2) applies to this faithful string C-marking. Its degree $N = m - 1 \geq 5$ is odd and its rank $N - 2 > N/2$. Every proper exception in that theorem has even degree. Thus T is the natural A_N or S_N. The centralizing permutation δ fixes y, so it fixes its entire H-orbit V_0 pointwise. It therefore preserves U_0, the only other nontrivial H-orbit. The centralizer of natural A_N or S_N in S_N is trivial for $N \geq 5$, so δ fixes U_0 as well. Together with b, x, D, these account for all points, contradicting oddness of δ.

If T is intransitive, Lemma 1 applies. Let p be its singleton in U_0 and $p' = p^\sigma$ the corresponding singleton in V_0. On the remaining two rows, H is the diagonal natural S_{m-2}, where $m - 2 \geq 4$. Its fixed points in E are precisely b, x, p, p'. On the two large rows its centralizer consists of the identity and the full matching that exchanges the rows: each separate natural centralizer is trivial, and the matching is the unique equivariant bijection. This exchange has $m - 2$ transpositions, an even number.

If $y = p'$, fixation of b, x, y forces δ to fix all four fixed points. Its remaining action is even, a contradiction. Otherwise y lies in a large row. Fixing y excludes the row exchange and makes δ fix both large rows pointwise. As δ is odd and fixes b, x, it must be exactly $(p\ p')$. But H fixes this pair pointwise, σ interchanges its two points, and e preserves it because e centralizes δ. Thus $B = \langle \sigma, e, H\rangle$ preserves $\{p, p'\}$, again contradicting transitivity. Both cases are impossible. □

Attribution and boundary. The critical-head and early-cut calculations remain acquired inputs. The complete L047 recovery, including its common theory, finite boundary, software and distinct reviews, is charged as a prior result of Cameron, Fernandes, Leemans and Mixer recovered in this project. Only its odd-degree consequence is applied; no L044 alternating rank induction is used here. The new argument uses the diagonal row action and the odd exterior involution to exclude two large blocks. Other block systems, opposite heads, the unshared endpoint and the full L044 recovery remain separate. No new finite census, formal theorem, whole-original or open-question credit is claimed.